

Question Number	Answer	Additional Guidance	Mark
2(a)	A description that includes four of the following - touching woodlouse in standardised manner (1) method of control of suitable variable (1) - repeat with different woodlice (1) one of these methods - repeated touches on (same) woodlouse (until no response seen) (1) - record the number of touches before no response seen (1) OR - measure time to curl/uncurl (record time taken and) number/frequency of touches (to curl/uncurl to standard state)	Accept eg glass rod / cotton bud / same force Accept temperature – AC room humidity /moisture – AC room light intensity – same intensity bulb / same distance woodlouse variable eg age – same length/ mass species – use a key Accept repeat with different woodlice	Exp (4)

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2(b)(i)	71 / 71.15/ 71.2	Not 71.1 / 71.0 / 71.20 Ignore minus sign	Cleri (1)

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2(b)(ii)	- axes correctly orientated with suitable axis labels and units and linear scale (1) - all points correctly plotted (1) - points (joined) with {labels/ key}(1)	eg. x axis: stimulus / sound y-axis: mean distance / cm (not MP3 if a bar graph)	Exp (3)

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2(b)(iii)	suitable precaution (1)	eg. Sound should not be too loud/time for fish to rest after testing/ do not test same fish several times; suitable size tank ignore release to the wild / control of light intensity	Exp (1)
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2(b)(iv)	Abiotic - temperature - sound factor eg intensity / duration of sound / how far away the sound is - light {intensity/wavelength} - size of tank - pH Biotic - Age / sex of fish	Accept pitch /intensity/amplitude/frequency Accept gender/mass	Exp (2)

Question Number	Answer	Additional Guidance	Mark
2(b)(v)	- variable with suitable control method described (1) - results are not valid / description of expected effect on the dependent variable (1)	Accept: Water temperature – use of thermostatic heater in tank Sound factors – sound of fixed duration eg 1s / from fixed distance eg 10m Light intensity – bulb of known wattage / at fixed distance Size of tank – same sized tank for each trial Age – fish hatched at same time Sex - select fish of known sex pH - buffer Description needs to be directional Accept other variables controlled and expected effect	Exp (2)

(Total for Question 2 = 13 marks)

